

# Data Visualization Report

Project 4 — Data Analytics Internship

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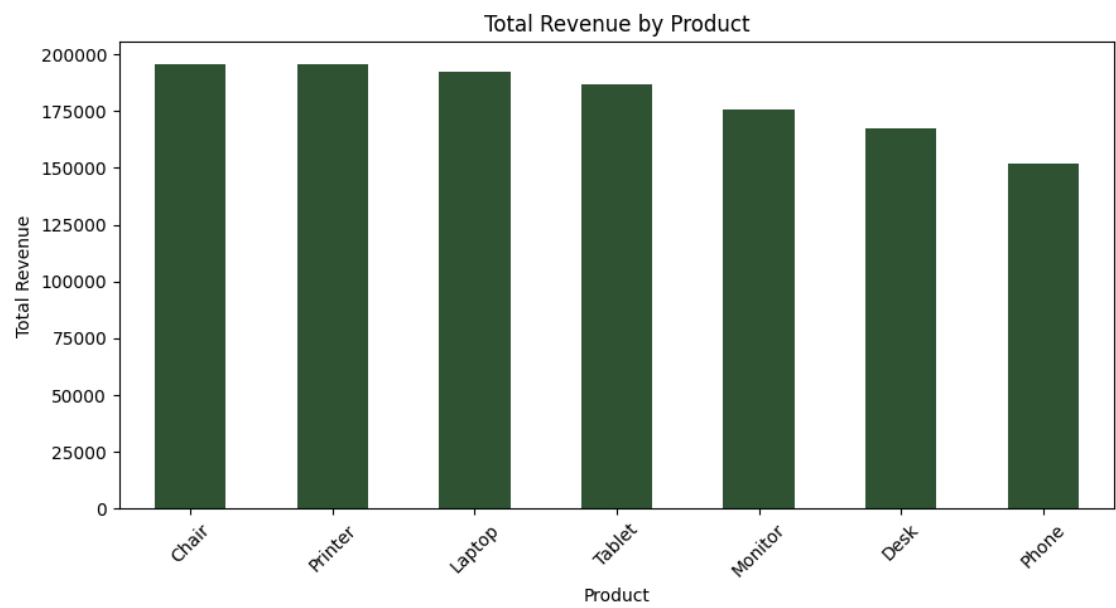
Prepared for: **DecodeLabs**

Track: Data Analytics

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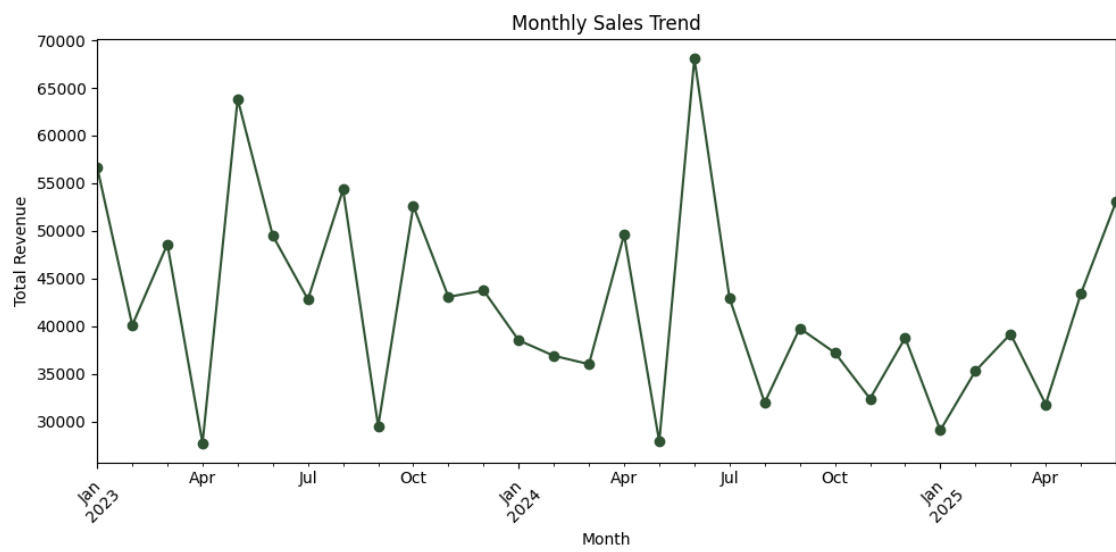
This report presents a visual analysis of the cleaned e-commerce order dataset (1,200 records), covering revenue distribution by product, monthly sales trends, payment method usage, and an executive-level Pareto analysis of revenue concentration.

## 1. Total Revenue by Product



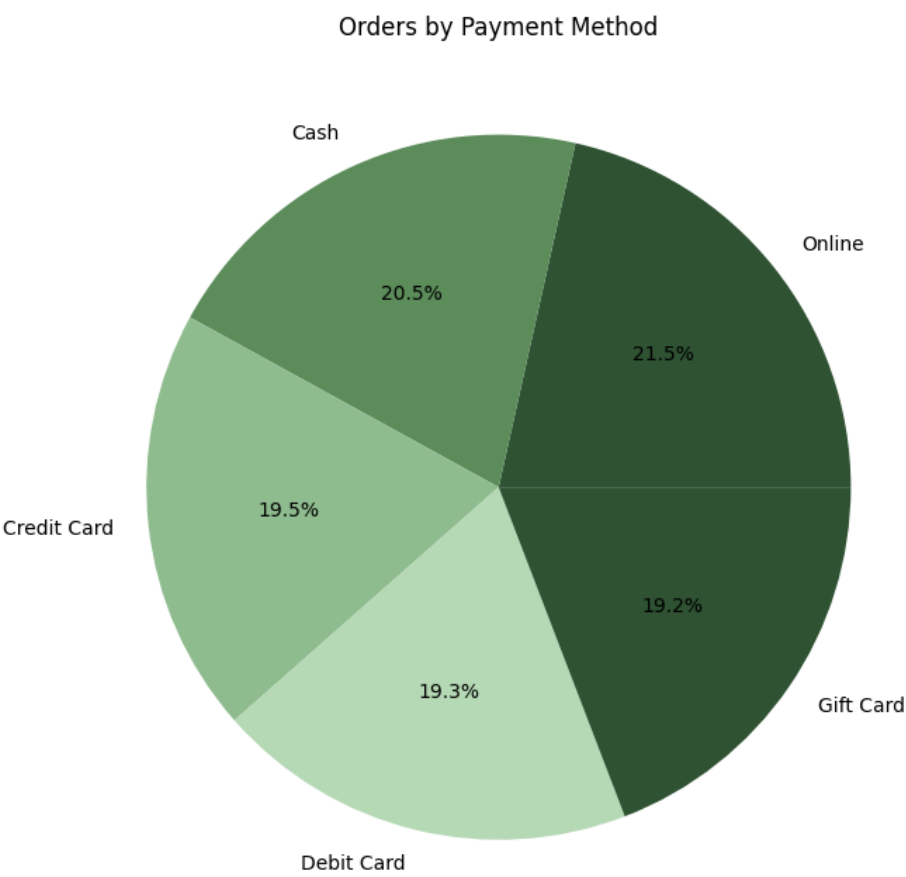
**Insight:** Chair and Printer generate the highest total revenue, while Phone contributes the least. Revenue across products is fairly close in range, suggesting a broad, well-balanced product portfolio rather than reliance on a single best-seller.

## 2. Monthly Sales Trend



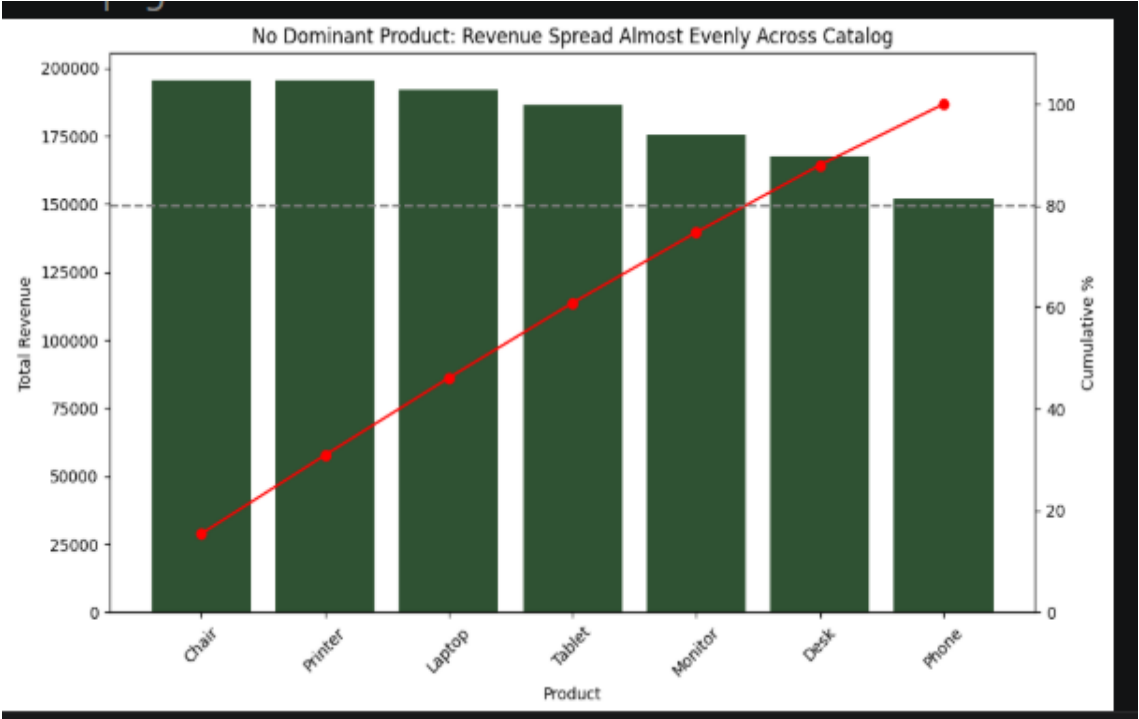
**Insight:** Monthly revenue fluctuates significantly, with a sharp peak around mid-2024 and periodic dips. This volatility suggests seasonal demand or promotional cycles that could benefit from deeper time-series investigation.

### 3. Orders by Payment Method



**Insight:** Payment method usage is almost evenly split across Online (21.5%), Cash (20.5%), Credit Card (19.5%), Debit Card (19.3%), and Gift Card (19.2%). No single method dominates, indicating customers are comfortable using a diverse set of payment options.

#### 4. Executive Insight: Revenue Concentration (Pareto Analysis)



**Insight:** Unlike a typical 80/20 pattern, revenue here is spread almost evenly across the product catalog — 6 of the 7 products are needed to reach 80% of total revenue. This means the business is not overly dependent on any single product, reducing risk but also indicating no single item is a standout revenue driver.

#### Summary of Key Findings

| Chart              | Key Finding                                              |
|--------------------|----------------------------------------------------------|
| Revenue by Product | Chair & Printer lead; revenue fairly balanced overall    |
| Monthly Trend      | High volatility with a notable mid-2024 peak             |
| Payment Method     | Evenly distributed across 5 methods (~19-22% each)       |
| Pareto Analysis    | No dominant product; 6/7 products needed for 80% revenue |

*Prepared as part of the Data Analytics Internship (Project 4: Data Visualization) at DecodeLabs.*